CHAPTER 02

Linear Programming Models

Turning a business problem into a model that can be solved

TEXTBOOK	CHAPTER	WEEK
Modern Management Science, 9th ed.	**Ch. 2, Examples 2-1 – 2-6**	**Week 2**

Management Science · Fall 2026 · Kitae Kim, School of Business, Korea Aerospace University

Content prepared by the instructor; slide layout and formatting assisted by Claude (Anthropic).

THIS HOUR

What we will cover

- What makes a model linear
- The three pillars of a linear programming model
- Formulation: the five steps
- Five worked cases
- Assumptions and limits of linear programming

Optimization

- Alternatives — what could we do?
- Preferences and constraints — which is better, and what is allowed?
- Optimal solution — the best alternative that satisfies every constraint

Alternatives and preferences both linear → a linear optimization model

QUICK CHECK

Which of these are linear?

A $Z = 2X + 3Y$

B $Y = 2X + 3X^2$

C $C = 8{,}000X_1 + 6{,}000X_2$

D $Y = \log(X)$

E $W = -2X + 3Y - Z$

F $Z = X / Y$

Is every variable standing alone, to the first power, times a number?

THE ANSWER

A sum of first-degree terms — nothing else

LINEAR

A $Z = 2X + 3Y$

C $C = 8{,}000X_1 + 6{,}000X_2$

E $W = -2X + 3Y - Z$

NOT LINEAR

B $Y = 2X + 3X^2$ — squared

D $Y = \log(X)$ — a function of X

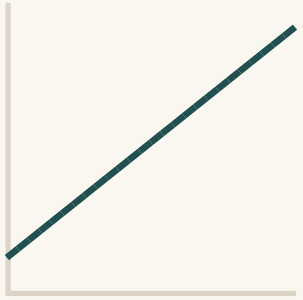
F $Z = X / Y$ — variables divided

F has no exponent — but dividing one variable by another is still nonlinear.

SEEN AS A PICTURE

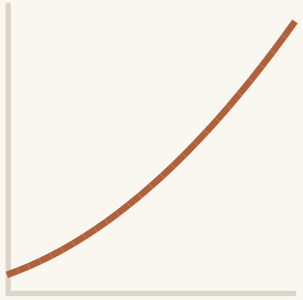
Linear means straight — literally

$Y = 2X + 3$



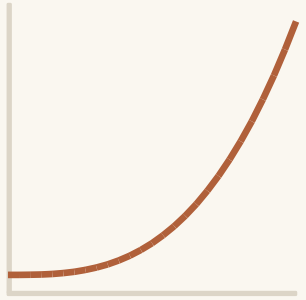
LINEAR

$Y = 2X + 3X^2$



NOT LINEAR

$Y = X^3$



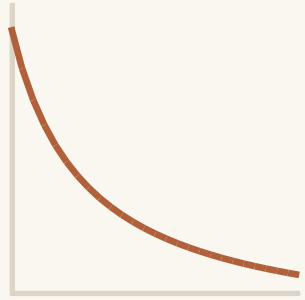
NOT LINEAR

$Y = \log(X)$



NOT LINEAR

$Y = 1 / X$



NOT LINEAR

A linear constraint has a straight boundary — so the optimum sits at a corner.

DEFINITION

Linear programming

- Building a linear optimization model of a decision, and using it to choose the best option
- Simplex method · Dantzig, 1947
- Interior-point method · Karmarkar, 1984
- The most widely used analytical tool in production, logistics and finance

“Programming” here means planning — the word predates computer programming.

Where linear programming is used

- Blending
- Product mix
- Transportation and distribution
- Production planning and inventory
- Financial engineering
- Staffing, routing and network design

Every LP model has three parts

- Decision variables — what are we deciding?
- Objective function — what are we trying to do?
- Constraints — what are the limits?

Miss one, or write a nonlinear line, and it is not a linear program.

Decision variables

- The quantities we actually control
- Ask: what number would you phone the factory and tell them?
- Ask: what is proportional to what?
- Always write the unit

Example 2-1

X_1 = cases of knives produced next week

X_2 = cases of forks produced next week

Objective function

- One linear expression — and exactly one
- Maximize: profit · revenue · output
- Minimize: cost · risk · time · waste
- Every term must land in the same unit

Example 2-1

$$\text{Maximize} \qquad 8{,}000X_1 + 6{,}000X_2 \qquad (\text{won})$$

won per case × cases = won

PILLAR 3

Constraints

- $\leq$ a ceiling — capacity, budget, supply
- $\geq$ a floor — requirement, demand, standard
- $=$ a balance — must be met exactly
- ≥ 0 non-negativity — never leave it out

Example 2-1

$$0.2X_1 + 0.4X_2 \leq 700$$
$$0.5X_1 + 0.25X_2 \leq 1{,}000$$
$$X_1 \geq 0,\ X_2 \geq 0$$

Formulation: five steps

1	**Read the situation**	mark every number, unit and keyword
2	**Set the decision variables**	what is proportional to what?
3	**Write the objective function**	one expression, maximize or minimize
4	**Write the constraints**	one line per limit, plus $X \geq 0$
5	**Assemble the model**	objective, subject to, constraints

STEP 1

Read the situation

- Mark every number
- Mark every unit — won/case, hours/unit, %
- Mark every keyword — available, at most, at least, exactly
- Find the goal sentence — usually the last one
- Ask: what is being decided, and by whom?

Most formulation errors are reading errors.

STEP 2

Set the decision variables

- Ask what is proportional to what
- **Follow the units**
  - “30,000 won per unit” → the coefficient is 30,000
- **Read the denominator**
  - won per case → cases are the variable
  - risk per 10,000 won → money is the variable

No unit, no variable.

STEP 3

Write the objective function

- Find the goal sentence → maximize or minimize
- One coefficient for each variable, added together
  - $8{,}000\ \text{won/case} \times X_1\ \text{cases} = 8{,}000X_1\ \text{won}$
  - $6{,}000\ \text{won/case} \times X_2\ \text{cases} = 6{,}000X_2\ \text{won}$
- Every term in the same unit — and exactly one objective

$$\text{Maximize} \qquad 8{,}000X_1 + 6{,}000X_2$$

STEP 4

Write the constraints

- at least · minimum → ≥
- at most · within · available → ≤
- exactly · requests → =
- then non-negativity → $X \geq 0$

Example 2-1, press process

$$0.2\ \text{h/case} \times X_1 + 0.4\ \text{h/case} \times X_2 \leq 700\ \text{h}$$

$$0.2X_1 + 0.4X_2 \leq 700$$

Hours on the left, hours on the right.

STEP 5

Assemble the model

X_1, X_2 = cases of knives, forks produced next week

$$\text{Maximize} \quad 8{,}000X_1 + 6{,}000X_2$$

$$\text{subject to} \quad 0.2X_1 + 0.4X_2 \le 700$$

$$0.5X_1 + 0.25X_2 \le 1{,}000$$

$$X_1 \ge 0, \; X_2 \ge 0$$

Variables defined · one objective · one line per limit · non-negativity · all linear

That is the whole method. Five cases follow — only the story changes.

Example 2-1

- Samyoung Industries makes table knives and forks, and exports all of it
- Both products pass through a press process and a polish process
- Set next week’s production so that total profit is as large as possible

	Knife	Fork	Available
Press (h / case)	0.2	0.4	700 h
Polish (h / case)	0.5	0.25	1,000 h
Profit (won / case)	8,000	6,000	—

Example 2-1

- **Variables**
  - X_1 = cases of knives, X_2 = cases of forks, per week
- **Objective**
  - Maximize $8{,}000X_1 + 6{,}000X_2$ (won)
- **Constraints**
  - $0.2X_1 + 0.4X_2 \le 700$ (press, hours)
  - $0.5X_1 + 0.25X_2 \le 1{,}000$ (polish, hours)
  - $X_1 \ge 0,\ X_2 \ge 0$

Example 2-1 — optimal solution

$$(X_1, X_2) = (1{,}500, 1{,}000)$$

Maximum weekly profit 18,000,000 won

- Press: $0.2(1{,}500) + 0.4(1{,}000) = 700$ hours — all of it
- Polish: $0.5(1{,}500) + 0.25(1{,}000) = 1{,}000$ hours — all of it
- Both constraints binding → both processes are bottlenecks

Example 2-2

- Build a daily menu from milk, tuna, bread and spinach
- Meet the daily requirement for vitamins A, C, D and iron, at least cost
- For taste: tuna ≥ 0.1 kg, bread ≥ 0.5 loaf

	Milk (L)	Tuna (kg)	Bread	Spinach (kg)	Need / day
Vitamin A (IU)	1,600	500	0	70,000	5,000
Vitamin C (mg)	10	0	0	140	30
Vitamin D (IU)	120	0	0	0	100
Iron (mg)	7	14	13	16	12
Price (won)	1,000	3,000	650	600	—

Example 2-2

- **Variables**
  - M, T, B, S = amount of milk, tuna, bread, spinach per day
- **Objective**
  - Minimize $M + 3T + 0.65B + 0.6S$ (thousand won)
- **Constraints**
  - $1{,}600M + 500T + 70{,}000S \geq 5{,}000$ $10M + 140S \geq 30$
  - $120M \geq 100$ $7M + 14T + 13B + 16S \geq 12$
  - $T \geq 0.1$, $B \geq 0.5$, $M, T, B, S \geq 0$

(M, T, B, S) = (0.83, 0.10, 0.50, 0.155) minimum cost ≈ 1,552 won per day

Example 2-3

- Company A has two plants and three warehouses
- Demand: 500, 2,000 and 900 units · Supply: ≤ 2,300 and ≤ 1,500
- Production cost: 150,000 won/unit at plant 1, 200,000 at plant 2
- Minimize production cost plus transport cost

Transport cost (10,000 won / unit)

	W 1	W 2	W 3
Plant 1	3	9	8
Plant 2	7	2	4

Example 2-3

- **Variables**
  - X_{ij} = units sent from plant i to warehouse j
- **Objective**
  - Minimize $18X_{11} + 24X_{12} + 23X_{13} + 27X_{21} + 22X_{22} + 24X_{23}$
- **Constraints**
  - $X_{11} + X_{21} = 500 \quad X_{12} + X_{22} = 2{,}000 \quad X_{13} + X_{23} = 900$
  - $X_{11} + X_{12} + X_{13} \le 2{,}300 \quad X_{21} + X_{22} + X_{23} \le 1{,}500$
  - $X_{ij} \ge 0$ for all i and j

(500, 500, 900, 0, 1,500, 0) minimum cost 747 million won

Example 2-5

- Company C makes three products on two types of machine
- Daily output required: 300, 700 and 400 units
- 100 machines of type 1 and 80 of type 2, each up to 8 hours a day
- Which machine should make which product, at least cost?

	Cost 1	Cost 2	Cost 3	Time 1	Time 2	Time 3
Machine 1	13	9	10	0.4	1.1	0.9
Machine 2	11	12	8	0.5	1.2	1.3

Cost in 10,000 won / unit · time in hours / unit

Example 2-5

- **Variables**
  - X_{ij} = units of product j made on machine type i
- **Objective**
  - Minimize $13X_{11} + 9X_{12} + 10X_{13} + 11X_{21} + 12X_{22} + 8X_{23}$
- **Constraints**
  - $0.4X_{11} + 1.1X_{12} + 0.9X_{13} \le 800 \qquad 0.5X_{21} + 1.2X_{22} + 1.3X_{23} \le 640$
  - $X_{11} + X_{21} = 300 \qquad X_{12} + X_{22} = 700 \qquad X_{13} + X_{23} = 400$
  - $X_{ij} \ge 0$ for all i and j

(0, 700, 23, 300, 0, 377) minimum cost ≈ 128.5 million won

Example 2-6

- M Asset has 20 billion won to invest across four stocks
- Expected return must be at least 1.8 billion won a year
- No more than 50% of the fund in any one stock
- Minimize the total risk index

Stock	A	B	C	D
Price (10,000 won / share)	10	5	8	4
Expected annual return (%)	12	8	3	10
Risk index per 10,000 won	0.1	0.07	0.01	0.08

Example 2-6

- **Variables**
  - A, B, C, D = amount invested in each stock, in units of 10,000 won
- **Objective**
  - Minimize $0.1A + 0.07B + 0.01C + 0.08D$
- **Constraints**
  - $A + B + C + D = 2{,}000{,}000$
  - $0.12A + 0.08B + 0.03C + 0.1D \geq 180{,}000$
  - $A, B, C, D \leq 1{,}000{,}000$ $A, B, C, D \geq 0$

A 10.0 bn · B 0 · C 5.71 bn · D 4.29 bn total risk index 140,000

LOOKING BACK

Five industries, one structure

Case	Decision variable	Objective	Constraints
Product mix	units of each product	max profit	≤ capacity
Diet	amount of each food	min cost	≥ requirement
Distribution	units on each route	min cost	= demand, ≤ supply
Scheduling	units on each machine	min cost	= output, ≤ hours
Portfolio	money in each asset	min risk	= fund, ≥ return

Name the variable, with its unit, and the model writes itself.

Assumptions and limits of LP

- Certainty — every coefficient is known, no probabilities
- Proportionality — double the input, double the output
- Additivity — the total is the sum of the parts
- Divisibility — fractions are allowed: 0.83 litres, or 2.4 aircraft

Each one is a claim about the world — and each one is sometimes false.

IN SUMMARY

Chapter 2

- Every LP has three parts: variables, one objective, constraints
- Formulation is a procedure — five steps, every time
- The keyword fixes the symbol: at least → ≥ · at most → ≤ · exactly → =
- Certainty, proportionality, additivity, divisibility — the price of linearity

Practice · formulate, do not solve, problems 2-1 to 2-5

Next week · Chapter 3 — solving an LP: the graphical method